

Evaluating User Acceptance using WebXR for an Augmented Reality Information System

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ABSTRACT

Augmented Reality has a long history and has seen major technical advantages in the last years. With *WebXR*, a new web standard, Mobile Augmented Reality (MAR) applications are now available in the web browser. With our work, we implemented an Augmented Reality Information System and conducted a case study to evaluate the user acceptance of such an application build with *WebXR*. Our results indicate that the user acceptance regarding web-based MAR applications for our specific use case seems to be given. With our proposed architecture we also lay the foundation for other AR information systems.

Index Terms: User Acceptance—Augmented Reality—WebXR—Information System

1 INTRODUCTION

With markerless tracking and registration and no need for additional apps, *WebXR* opens a wide variety of use cases. This paper deals with the question if this new API may fit the requirements of a real-time MAR information system, to facilitate the work or use of a FabLab. In trying to answer this question, we have conducted a case study by evaluating four different exemplary scenarios a FabLab user or employee may encounter. FabLabs are workshop-like spaces equipped with a variety of, often computer-controlled, machines for the production of all kinds of objects.

2 MOTIVATION

Running a FabLab involves many staff-intensive, repetitive tasks, especially in assisting visitors and users of the FabLab. Many of these tasks, such as assisting with orientation and use of the equipment in the FabLab, would benefit greatly from automation. As *WebXR* is a de facto web standard now, the start of an AR application has become as easy as visiting a website. Before *WebXR*, browser-based AR experiences heavily depended on special Markers¹ for tracking. Alike, other research of the last years has shown that MAR applications can be applied in industry and maintenance applications [4].

3 RELATED WORK

Chatzopoulos et al. [1] as well as Dini et al. [2] did extensive literature research and identified several suitable fields of MAR applications. Both identified maintenance and inspections as one of the major job-related applications. Tang et al. [5] showed that assistive MAR applications can enhance assembly and maintenance jobs and reduce assembly error by 82%. Dirin et al. [3], amongst other, identified mental, technical and User Interface challenges and

recommended UX design practices for MAR applications (especially information systems), our survey is focused on these challenges.

4 METHODOLOGY

To assess the suitability of *WebXR* for a real-time MAR information system, we conducted a case study in our FabLab of the Ruhr-West University of Applied Sciences in Bottrop. The HRW FabLab currently has 23 employees, seven of whom participated in our case study. To test the user acceptance, we created four typical scenarios an employee or user of the FabLab could encounter. The user feedback was collected in a qualitative survey. The following section describes these steps in detail.

4.1 Software Architecture

Our system consists of a web application that typically runs on a smartphone or tablet. The backend provides an API for real-time communication with the smartphone. Besides that, it provides a REST API for both updates of the FabLab infrastructure (e.g. occupation status of a machine) and updates of the virtual scene by a content author. The backend and mobile application are implemented in a generic fashion, so they are capable of implementing applications for a wide variety of domains.

4.1.1 Mobile Application

The mobile application, hereafter called *app*, is a single page web app. The user interface was built with *React*². The *WebXR* specific part was built with the documentation and guidelines³ of the *Immersive Web Working Group*⁴. We did choose *three.js*⁵ as 3D engine for the rendering of the 3D objects. The image output of *three.js* is directly written to the framebuffer, provided by the *WebXR* API, and thereby superimposes the camera image. For our app, we used the "immersive-ar" session type of the *WebXR* API. The API delivers a pose of the mobile device in the room after successful registration, which we then used to control the camera in the *three.js* scene.

4.1.2 Backend

The backend, hereafter called *CMS*, is a *node.js*⁶ application in TypeScript⁷. We used *Express*⁸ as web framework. For storage of the scene objects, 3D models and materials, we used *mongodb*⁹, as it enabled us to store the used 3D models in a JSON file format called *glTF*¹⁰. The CMS provides two APIs. A REST API for the authoring of the scene, e.g. adding/removing of scene objects, transformations of scene objects (position, size, rotation), or uploading of 3D models. A real-time API via WebSockets for the app to update the 3D scene, when the CMS' scene representation is changed via the REST API.

²<https://reactjs.org/>

³<https://github.com/immersive-web/webxr>

⁴<https://www.w3.org/immersive-web/>

⁵<https://threejs.org/>

⁶<https://nodejs.org>

⁷<https://www.typescriptlang.org/>

⁸<https://expressjs.com>

⁹<https://www.mongodb.com/de>

¹⁰<https://www.khronos.org/glTF/>

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¹<https://gitlab.hs-ruhrwest.de/fablabar>

4.2 Case Study

The aim to test the user acceptance of *WebXR* in the FabLab area was achieved by using two methods. First, each participant had to perform four different scenarios with the help of the *WebXR* application we created. After completing the four scenarios, each participant had to fill out a survey in which they had to describe suggestions for improvement, use cases, and their general satisfaction with the app.

4.2.1 Predefined Scenarios

To test our application, four different scenarios were created. The scenarios were everyday tasks that are common in a FabLab: The Availability of a Machine, The Lending of a Drilling Machine, Maintenance Mode and Navigation in an Immutable Environment.

5 EVALUATION

In the following section we discuss the results of the survey. We also discuss the limitations we have faced in our work with *WebXR* and our chosen architecture.

5.1 Survey

To record the results of our scenarios, a web-based survey was created. The survey contained 20 questions and was completed by all seven participants. As shown in Table 1, the results of our study were consistently positive. None of the FabLab employees had previously used an Augmented Reality application in a working environment. All seven participants still found our *WebXR* application easy to use. Six of the interviewees reported that they consider the app as a simplification in their daily work. There were several reasons for this. First of all, it is noteworthy that four of the seven participants praised the smooth performance of our *WebXR* application independently of the survey. Three of the seven participants also commented that not having to install an app is very beneficial because you do not have to trust any application and can save storage space on your phone. In five cases, participants reported that the application could almost automate the interaction with FabLab visitors. For example, FabLab visitors would be able to independently identify free 3D printers and would no longer have to be guided to the location of the respective device when borrowing items. The application could do this automatically. Another positive aspect is the maintenance mode. With the help of our application, a maintenance team could immediately recognize all defective devices. This way, a maintenance can be started directly and no FabLab employee is needed to instruct the maintenance team.

Table 1: Evaluation of Case Study (n=7)

Question	Yes	No
Ever used an AR-Application in working environment	0%	100%
Handling of the application was easy	100%	0%
Would make my work easier	85.71%	14.29%

There were however also some suggestions for improvement. One problem was of the loss of tracking, it can be noted that the *WebXR* framework has problems when the premises have large empty areas combined with a floor with very little contrast. The AR scene is calibrated in combination with a hit test during initialization. If there are too few reference points, *WebXR* has problems calculating the real world position of the device. The main point of criticism is the currently limited compatibility of *WebXR* on the test subjects' mobile phones. Five out of seven participants had to use a smartphone provided by us, because their device was not compatible with *WebXR*. However, this is largely due to the development stage of *WebXR*.

5.2 Limitations

In the development process, we encountered different limitations regarding the *WebXR* API in general but also the architecture we chose to implement and the amount of supported devices.

5.2.1 WebXR

WebXR is a relatively new browser API, hence the feature set is yet relatively limited. The primary features needed by our *MAR* application are overall working as expected. The registration and tracking are working well under certain environmental conditions e.g. good lighting conditions and visually feature-rich environments. Also, the *hit point* API provided by *WebXR* works well under these conditions. The limitations we encountered here were to be expected from a system that primarily uses visual features for registration and tracking. However *WebXR* restricts the access to the camera's video stream for security reasons, which is totally reasonable. However, it also limits the further processing of the environment for certain use-cases. It is for example currently not possible to search for additional markers in the video stream. This would have helped us a lot, especially for the initial calibration and more precise positioning of content in the near field.

6 CONCLUSION

We found that *WebXR* can fit the requirements for a *MAR* information system. Through the survey, we found that our application met nearly all of the challenges described by Dirin et al. [3]. Additionally, also unexperienced users found the usage of our application quite easy. Nevertheless, we also met some hurdles. Only a fraction of the participants with an Android smartphone could use their own devices. On Apple devices *WebXR* is not yet supported. We also encountered some problems regarding the *Spatial Correspondence* and *Tolerance to movement*. Users were able to see objects through walls, which lead to misunderstandings of the environment. Sometimes the tracking of some participants got lost on fast device movement. This is seemingly due to the markerless nature of the tracking, where an environment with high contrast features (edges and patterns) is needed.

7 OUTLOOK

In a next iteration of our study, a test in different environments would be necessary to further support our findings. Furthermore, the *WebXR* API forbids the processing of the camera image for security reasons, which prevents the usages of markers for special applications. A solution for this would be favorable. The further development of *WebXR* will also show how many devices really benefit from Augmented Reality through *WebXR* and whether iOS will open their system to the new web standard. Additionally, the investigation of different applications would be needed to further substantiate the suitability of *WebXR* for *MAR* information systems.

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